

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

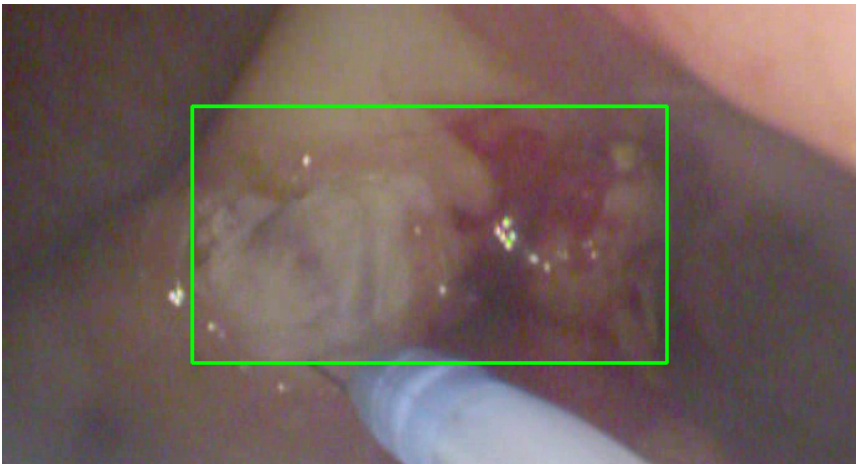
Video ID:	cafa734d-74ea-48f2-b58a-3f3151a99177
Total Frames:	3006
Frame Rate:	25.0 fps (assumed)
Duration:	120.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	86	✓
RT-DETR Detections	1157	✓
MedSAM2 Detections	1154	✓
Consensus (3-Model Agreement)	8	✓
Risk Classification	Count	Percentage
HIGH RISK	1	12.5%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	8	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — NORMAL_MUCOSA [BASELINE] (Tracked across 80 frames)

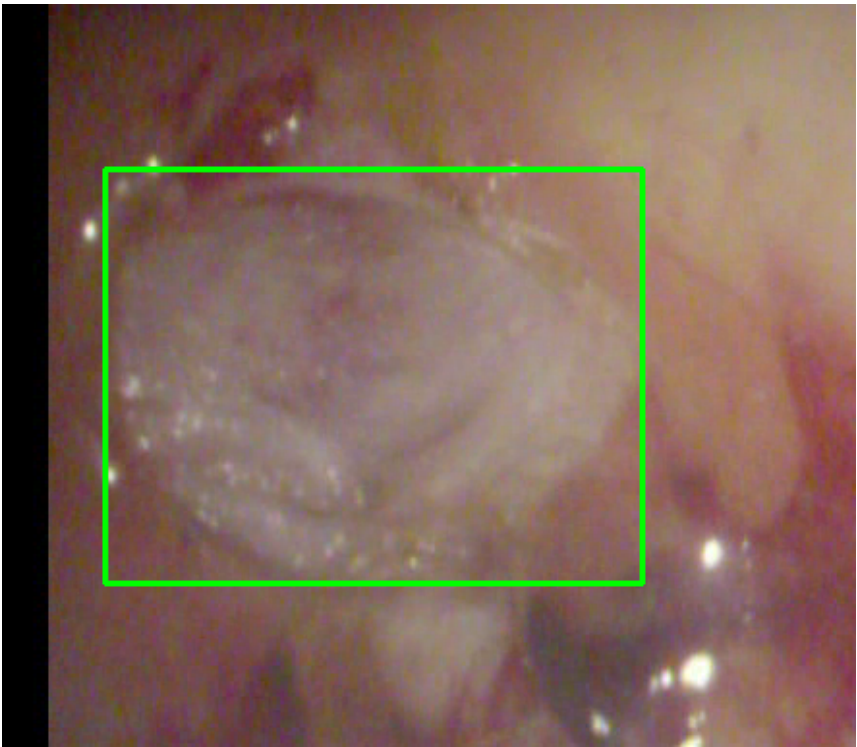


Feature	Value	Interpretation
Frame Index	810	Frame 810 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	NORMAL_MUCOSA	Type confidence: 72.8%
Redness Score	0.124	Color-based vascularization
Relative Radius	13.6%	Polyp size as % of frame diagonal
Texture Score	0.019	Surface morphology
Vessel Visibility	0.211	Vascularization indicator
Risk Tier	BASELINE	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	64.1%	YOLO / RT-DETR / track confidence

Type Assessment: Normal mucosal pattern — no polyp morphology identified.

Insufficient evidence for a stable risk prediction - Further evaluation recommended

POLYP #2 — NORMAL_MUCOSA [BASELINE] (Tracked across 109 frames)



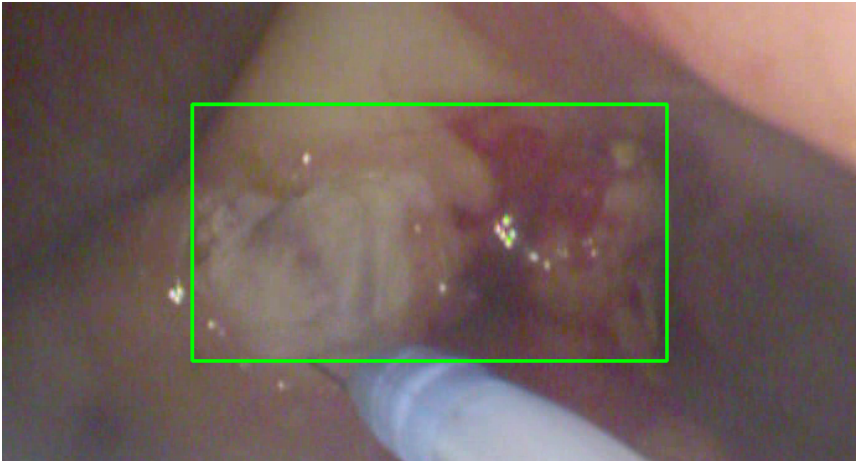
Feature	Value	Interpretation
Frame Index	2494	Frame 2494 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 Unavailable for this video)
POLYP TYPE	NORMAL_MUCOSA	Type confidence: 64.8%
Redness Score	0.106	Color-based vascularization
Relative Radius	23.2%	Polyp size as % of frame diagonal
Texture Score	0.015	Surface morphology
Vessel Visibility	0.036	Vascularization indicator
Risk Tier	BASELINE	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	62.7%	YOLO / RT-DETR / track confidence

Type Assessment: Normal mucosal pattern — no polyp morphology identified.

Insufficient evidence for a stable risk prediction - Further evaluation recommended

POLYP #3 — NORMAL_MUCOSA [BASELINE] (Tracked across 80 frames)



Feature	Value	Interpretation
Frame Index	810	Frame 810 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	NORMAL_MUC OSA	Type confidence: 72.8%
Redness Score	0.124	Color-based vascularization
Relative Radius	13.6%	Polyp size as % of frame diagonal
Texture Score	0.019	Surface morphology
Vessel Visibility	0.211	Vascularization indicator
Risk Tier	BASELINE	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	59.4%	YOLO / RT-DETR / track confidence

Type Assessment: Normal mucosal pattern — no polyp morphology identified.

Insufficient evidence for a stable risk prediction - Further evaluation recommended

POLYP #4 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 24 frames)

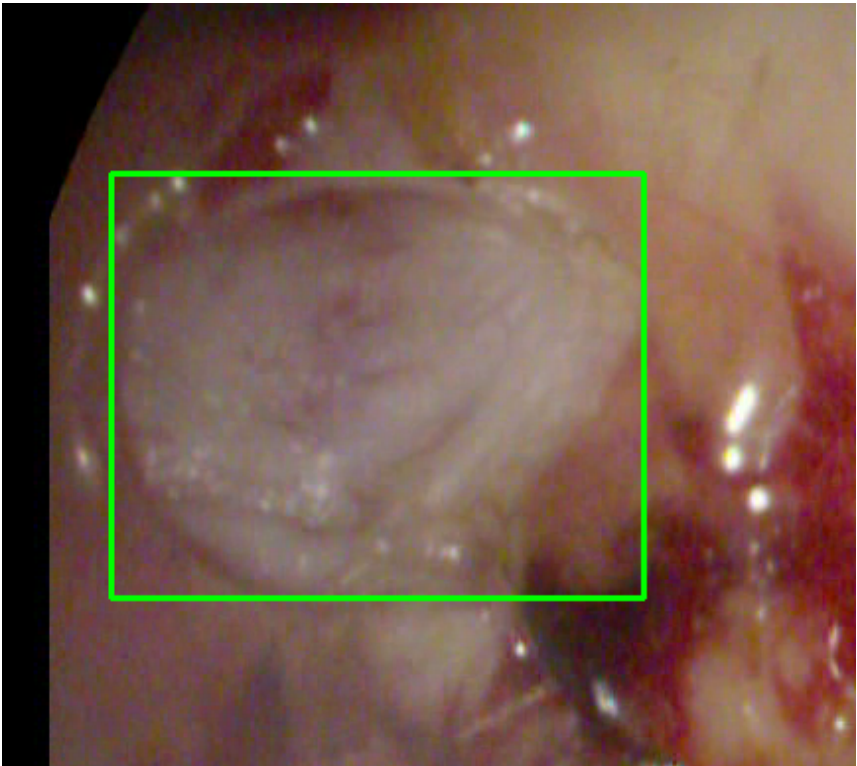


Feature	Value	Interpretation
Frame Index	2568	Frame 2568 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.213	Color-based vascularization
Relative Radius	15.4%	Polyp size as % of frame diagonal
Texture Score	0.014	Surface morphology
Vessel Visibility	0.634	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	65.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #5 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 44 frames)



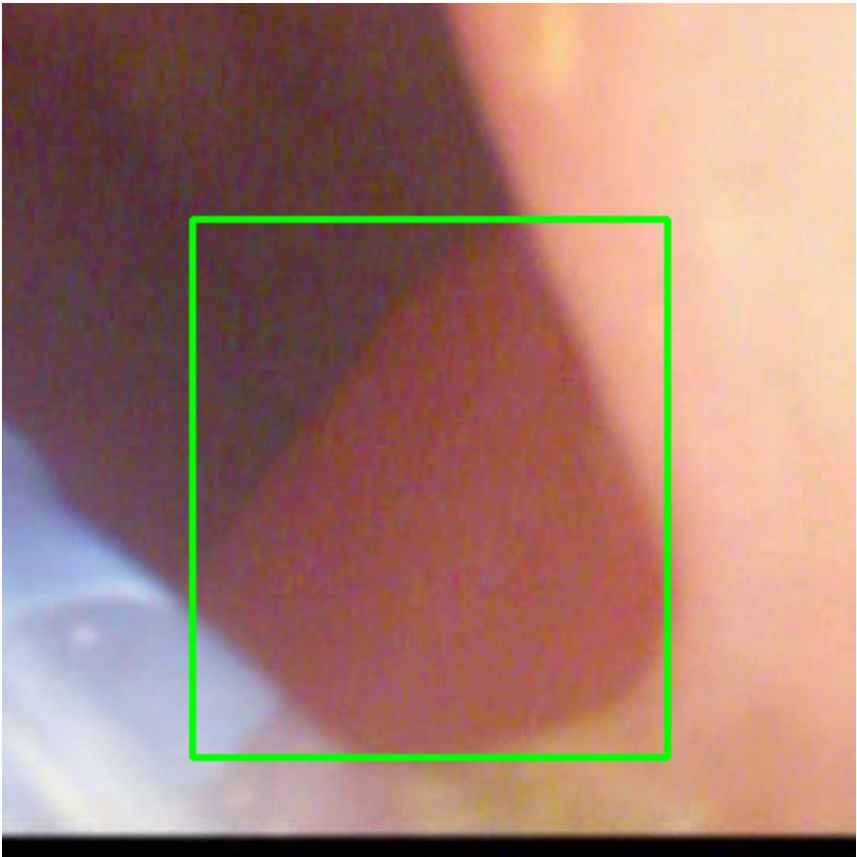
Feature	Value	Interpretation
Frame Index	2568	Frame 2568 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.213	Color-based vascularization
Relative Radius	15.4%	Polyp size as % of frame diagonal
Texture Score	0.014	Surface morphology
Vessel Visibility	0.634	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	ADENOMATOUS POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	67.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #6 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 36 frames)



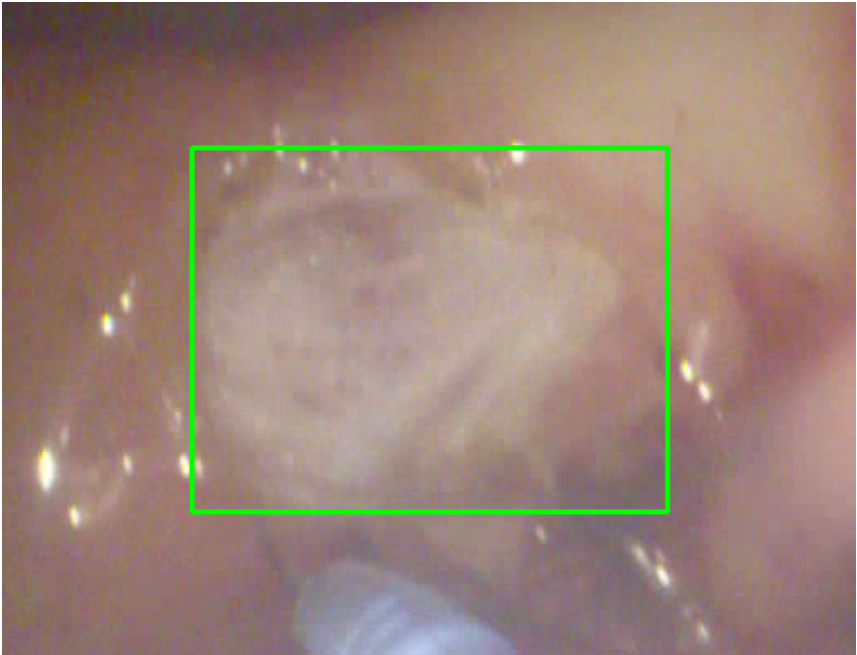
Feature	Value	Interpretation
Frame Index	2928	Frame 2928 of 3006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.202	Color-based vascularization

Relative Radius	17.2%	Polyp size as % of frame diagonal
Texture Score	0.012	Surface morphology
Vessel Visibility	0.728	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	74.0%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

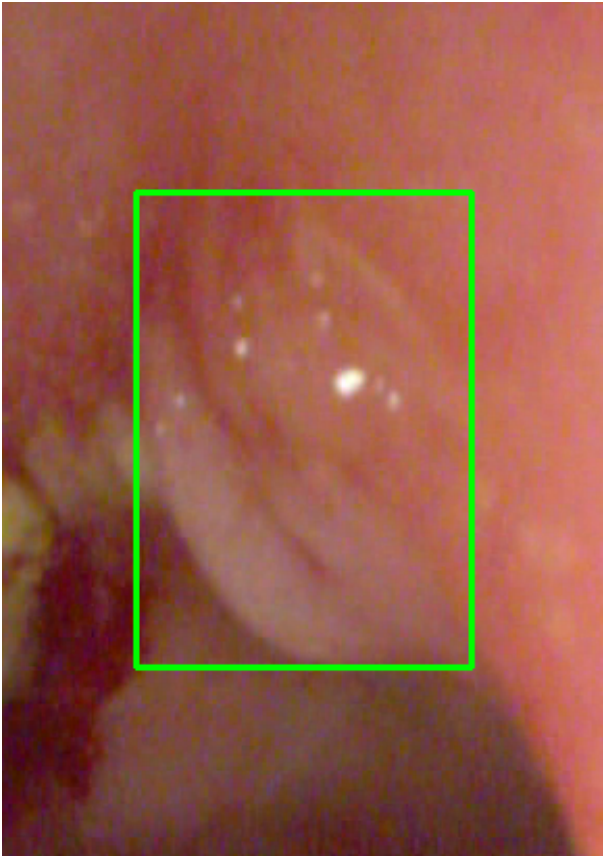
POLYP #7 — LIFTED_POLYP [LOW_RISK] (Tracked across 33 frames)



Feature	Value	Interpretation
Frame Index	1716	Frame 1716 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 55.8%
Redness Score	0.044	Color-based vascularization
Relative Radius	26.3%	Polyp size as % of frame diagonal
Texture Score	0.024	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	63.7%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

POLYP #8 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 44 frames)



Feature	Value	Interpretation
Frame Index	2568	Frame 2568 of 3006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.213	Color-based vascularization
Relative Radius	15.4%	Polyp size as % of frame diagonal
Texture Score	0.014	Surface morphology
Vessel Visibility	0.634	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	64.1%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 8 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 1 (12.5%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.